

CHECK WHAT YOU LEARNT

Check Yourself - Unit 4**A. Choose the correct answer**

- The DHT11 measures
(a) only temperature (b) temperature and humidity (c) only humidity (d) light and heat
- A DHT11 can give a fresh reading roughly once every
(a) 2 milliseconds (b) 2 seconds (c) 2 minutes (d) 2 hours
- A library is
(a) a type of sensor (b) ready-made code you can use (c) a wiring diagram (d) an error
- isnan(t) checks whether
(a) t is too hot (b) the sensor failed to answer (c) t is negative (d) the LED is on
- Humidity is measured in
(a) degrees (b) percent (c) ohms (d) volts
- Using different switch-on and switch-off points is called
(a) calibration (b) hysteresis (c) mapping (d) debugging

B. Fill in the blanks

- The DHT11 sends its reading out of the _____ pin.
- A temperature value should be stored in a variable of type _____.
- Adafruit's DHT library is installed from Tools > Manage _____.
- A one-degree gap between switching points stops the LED from _____.

C. Answer in one or two lines

- Write the cold and hot limits you chose, and the data that led you to them.
- Why is it a mistake to ask a DHT11 for a reading every 100 milliseconds?
- Explain hysteresis to a Class 6 student in two sentences.
- Why did we write allLedsOff() instead of repeating three lines each time?

UNIT 5

Final Project: Smart Rain Alarm System

Sessions 12, 13, 14 and 15

Your final machine uses two senses at once. It detects not just whether it is raining, but how heavily - and it stays sensibly quiet at night. This is the project you will demonstrate.

By the end of this unit you will be able to:

- Explain how a rain drop sensor turns water into a number.
- Combine two analog sensors in one decision.
- Design alert patterns that tell severity by sound alone.
- Calibrate a machine using real water and defend your numbers.

SESSION 12

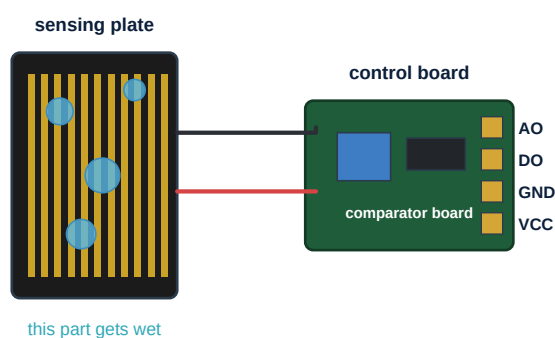
one class

How a Rain Sensor Works

Today you will

- Understand the sensing plate and the control board.
- Learn why AO is more useful than DO for this project.
- Agree the team roles and plan the build.

The black plate is covered with two gold tracks that weave between each other but never touch. Dry, no electricity can cross the gap. When water lands on the plate it bridges those tracks, because rainwater conducts electricity slightly. A little water makes a poor bridge; a lot of water makes a good one. That difference is exactly what lets your machine judge how heavy the rain is.

**Two boards, one sensor****AO - analog out**

a number from 0 to 1023: how wet the plate is

DO - digital out

just wet or dry, set by the small blue screw

In this project you use AO, because you want to know how heavy the rain is, not merely whether it is raining.

Figure 5.1 - The rain sensor comes as two pieces joined by two wires

Notice the choice you are making. The DO pin would tell you only *raining* or *not raining* - a Class 6 answer. The AO pin gives you the full range, so your alarm can distinguish a drizzle from a downpour. That is a Class 7 answer, and it is the whole reason this project is worth building.

Watch Out

Only the plate may get wet. Keep the green control board, the breadboard and the Arduino well away from the water, and mount the plate at a slight angle so drops run off instead of pooling. A pool of water reads as heavy rain long after the rain has stopped.

Role	Responsible for	Name
Builder	Mounting the plate at an angle and keeping water away from the electronics	_____
Wiring engineer	Every connection, checked twice before power on	_____
Programmer	Typing, uploading and adjusting the threshold values	_____
Recorder	The notebook: readings, faults, and what fixed them	_____

SESSION 13

one class

Building the Alarm**Today you will**

- Wire the rain sensor, the LDR, three LEDs and the buzzer.
- Understand why the LDR is included at all.
- Get a first working version running.

Smart rain alarm - connections

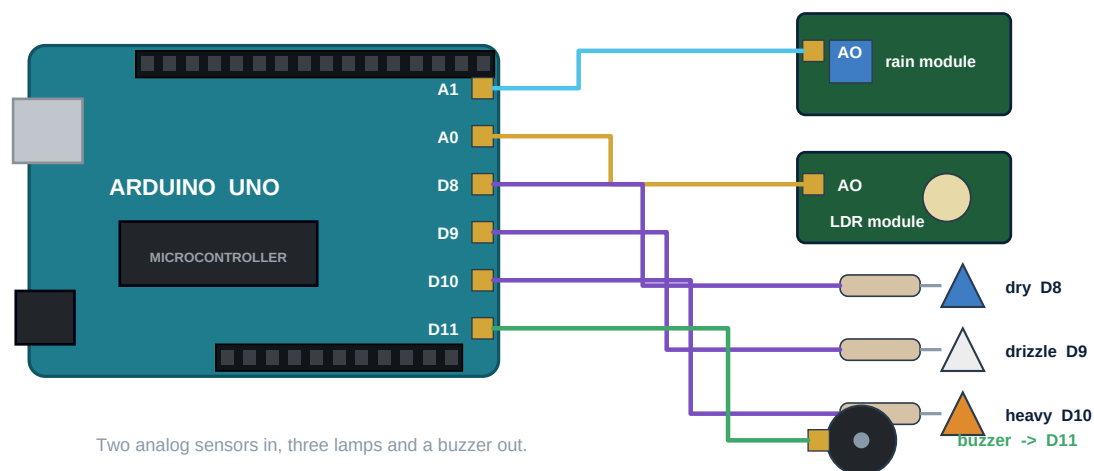


Figure 5.2 - Two senses working together

Component pin	Arduino pin	Why
Rain module AO	A1	How wet the plate is, 0 to 1023
Rain module VCC / GND	5V / GND	Power
LDR module AO	A0	How bright it is outside
LDR module VCC / GND	5V / GND	Power
Blue LED long leg	D8 through 220 ohm	Dry
White LED long leg	D9 through 220 ohm	Drizzle
Orange LED long leg	D10 through 220 ohm	Heavy rain
Buzzer +	D11	The alarm itself

Why the LDR? Because a good machine is considerate. An alarm that shrieks at three in the morning gets unplugged, and an unplugged alarm protects nobody. The LDR lets your system show the lamps silently at night while still sounding the buzzer in daylight, when somebody might actually act on the warning.

Did You Know

This idea has a name in engineering: **context awareness**. A phone that stays silent during a calendar meeting is doing exactly what your rain alarm does - using a second piece of information to decide how loudly to behave.

SESSION 14

one class

Calibration and Alarm Patterns**Today you will**

- Measure real dry, drizzle and heavy readings with water.
- Choose and justify your two thresholds.
- Give each level its own buzzer pattern.

Calibration means tuning a machine to the world it actually lives in. Your plate, your water and your room are not ours, so our numbers are only a starting point. Use a dropper or a wet finger and fill this table honestly before you set any threshold.

Condition of the plate	Reading on A1	What it should mean
Completely dry	_____	DRY
Three or four drops	_____	DRIZZLE
Sprayed all over	_____	HEAVY
Wiped dry again	_____	back to DRY
drizzleLevel you chose	_____	boundary 1
heavyLevel you chose	_____	boundary 2

Watch Out

Some rain modules count **up** as they get wetter and others count **down**. Find out which yours does before you write a single if statement. If your alarm behaves backwards, do not rewire anything - swap the > signs for <.

SmartRainAlarm.ino

```
int rainPin = A1;
int ldrPin  = A0;
int dryLed = 8, drizzleLed = 9, heavyLed = 10;
int buzzer = 11;

int drizzleLevel = 400;  // change these after your own testing
int heavyLevel   = 700;
int nightLevel   = 300;  // below this, treat it as night

void setup() {
  pinMode(dryLed, OUTPUT);
  pinMode(drizzleLed, OUTPUT);
  pinMode(heavyLed, OUTPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}

void allLedsOff() {
  digitalWrite(dryLed, LOW);
  digitalWrite(drizzleLed, LOW);
  digitalWrite(heavyLed, LOW);
}

void beep(int times, int gap) {  // our own alarm pattern
  for (int i = 0; i < times; i++) {
    digitalWrite(buzzer, HIGH);
    delay(120);
    digitalWrite(buzzer, LOW);
    delay(gap);
  }
}

void loop() {
  int wet  = analogRead(rainPin);
  int light = analogRead(ldrPin);
  boolean daytime = (light > nightLevel);

  Serial.print("wet: ");
  Serial.print(wet);
  Serial.print("  light: ");
  Serial.println(light);

  allLedsOff();

  if (wet < drizzleLevel) {
```

```

    digitalWrite(dryLed, HIGH);           // no rain, stay silent
  } else if (wet < heavyLevel) {
    digitalWrite(drizzleLed, HIGH);
    if (daytime) { beep(1, 800); }       // one slow beep
  } else {
    digitalWrite(heavyLed, HIGH);
    if (daytime) { beep(4, 120); }      // fast urgent beeps
  }

  delay(400);
}

```

The instruction `beep(times, gap)` is one you wrote yourself, and it takes two pieces of information: how many beeps, and how long to pause between them. That is what makes drizzle and downpour sound genuinely different. Somebody in the next room can tell how hard it is raining without looking at your machine at all - and that is good design.

SESSION 15

one class

Testing and Your Demonstration

Today you will

- Find and fix the faults in your own machine.
- Pass every acceptance test with real water.
- Present your project in three minutes.

Problem you may see	Most likely cause	What to do
Alarm stays on after the plate is dried	Water pooling between the tracks	Mount the plate at a steeper angle and wipe it properly
Readings jump about wildly	A loose jumper on A1, or a wet control board	Reseat the wire; keep the green board dry
Buzzer never sounds	The LDR thinks it is night	Print light and check it against nightLevel
Alarm behaves backwards	Your module counts down as it gets wetter	Swap the > signs for < in the ladder
Nothing works at all	Missing shared GND	One GND wire must reach every module

Acceptance test	Result	Tick
Dry plate: blue LED on, buzzer silent	_____	■
A few drops: white LED on, one slow beep	_____	■
Sprayed plate: orange LED on, fast beeps	_____	■
Plate wiped dry: returns to blue within 5 seconds	_____	■
Sensor covered to simulate night: lamps still work, buzzer silent	_____	■

15.1 Your Three-Minute Demonstration

- The problem** - one sentence on what your machine solves.
- How it senses** - hold up the plate and explain how water bridges the gold tracks.
- How it decides** - show the two threshold numbers and say where they came from.
- How it acts** - run it live with a water spray. Let the beeps tell the story.
- The clever bit** - explain why the LDR is there. This is what makes your project stand out.

6 What went wrong - name one fault you found and fixed.

Challenge

Add the SG90 servo as a model window shutter or clothesline cover. Wire its orange signal wire to D6, include the Servo library, and use `shutter.write(90);` to close it when the reading crosses `heavyLevel` and `shutter.write(0);` to reopen it once the plate is dry.

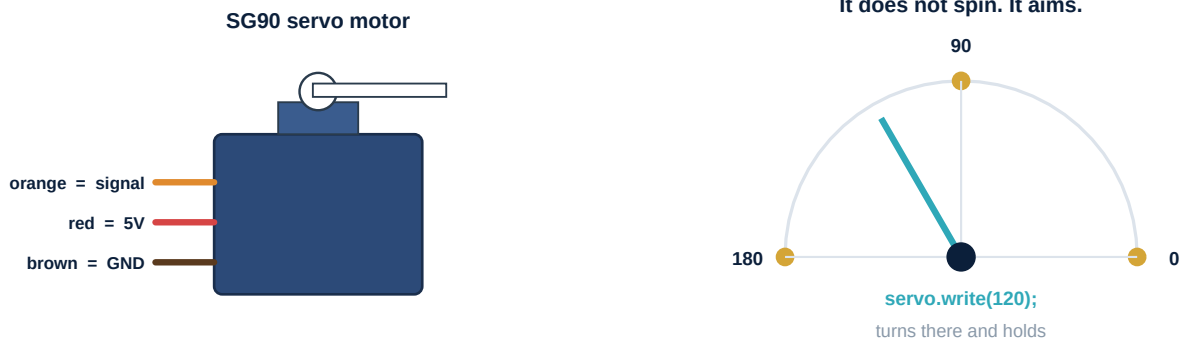


Figure 5.3 - A servo turns to an exact angle between 0 and 180 degrees

New word	What it means
Conductivity	How easily electricity passes through something. Water conducts a little.
Threshold	The number at which a decision changes from one answer to the other.
Calibration	Adjusting a machine to suit the real place it works in.
Context awareness	Using a second piece of information to decide how to behave.